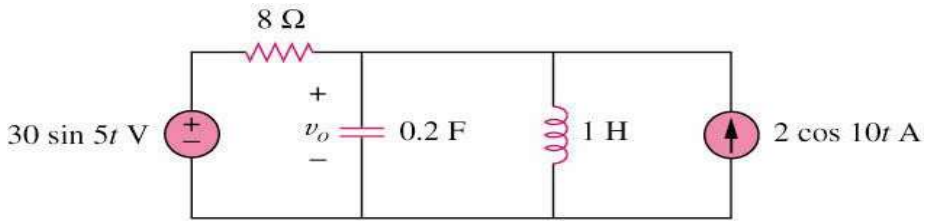
ELEC 226 HOMEWORK-1
KEY

Q-1) Calculate v_0 voltage in the circuit by using Superposition Theorem. (Use the given notation for v_0 voltage).



$Vs = 30 * e^{i*(-90)}$
$|Vs| = 30 \quad \alpha = -90\, deg$
$R = 8 \quad C = 0.2 \quad L = 1 \quad w1 = 5 \quad w2 = 10$
$w1 = 5\ için\ devrede\ sadece\ Vs\ var$
Akım kaynağı kısa devredir.

$$Zc1 = \frac{1}{i * w1 * C} = -i \qquad ZL1 = i * w1 * L = 5i$$

$$Reş1 = R + \frac{Zc1 * ZL1}{Zc1 + ZL1} = 8 - 1.25i$$

$$Vo1 = Zc1 * \frac{Vs}{Reş1} * \frac{ZL1}{Zc1 + ZL1}$$

$Vo1 = -4.576 - 0.715i$
$|Vo1| = 4.631 \quad \beta 1 = -171.119\, deg$

$w2 = 10\ için\ devrede\ sadece\ Is\ var$
gerilim kaynağı kısa devredir
$Is = 2 * e^{i*(0)} \quad |Is| = 2$

$$Zc2 = \frac{1}{i * w2 * C} = -0.5i$$

$ZL2 = i * w2 * L = 10i$

$$\frac{1}{Reş2} = \frac{1}{R} + \frac{1}{Zc2} + \frac{1}{ZL2}$$

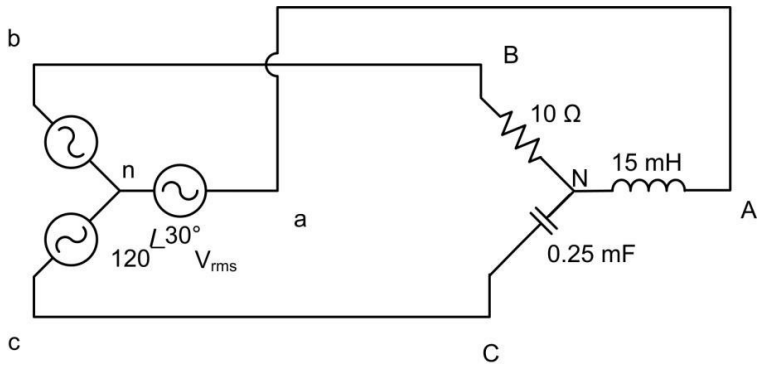
$Reş2 = 0.034 - 0.524i$
$Vo2 = Reş2 * |Is| = 0.069 - 1.048i$
$|Vo2| = 1.05 \qquad \beta 2 = -86.236\, deg$

$$Vo = Vo1 + Vo2 = 4.631\cos(5t - 171) + 1.05\cos(10t - 86.4)$$

Q-2) For the 3-ph. system given below, assume a balanced sourcewith a positive phase sequence.If the operating frequency is 50 Hz

a) Find the magnitude of V_{AN}, V_{BN}, V_{CN} voltages

b) Find the total complex power suppiled by the sources.



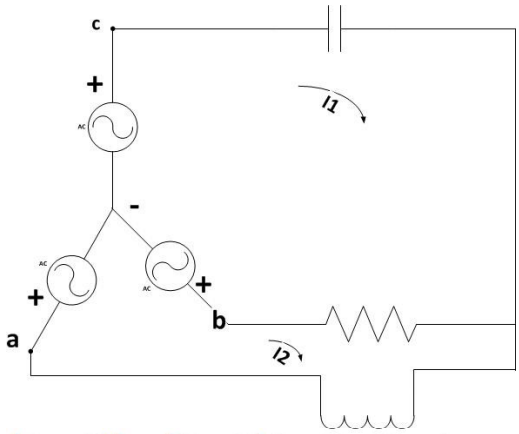
$Va = 120 * e^{i*(30)}$ $Vb = 120 * e^{i*(150)}$

$Vc = 120 * e^{i*(270}$ $f = 50$ $R = 10$

$L = 15 * 10^{-3}$ $C = 0.25 * 10^{-3}$

$ZL = i * 2 * \Pi * f * L = 4.715i$ $Zc = \frac{1}{i * w * C}$

$Zc = -12.732i$



$(Va - Vb) + (I2 - I1)R + I2 * ZL = 0$

$(Vb - Vc) + I1 * Zc + (I1 - I2) * R = 0$

$I1 * R - I2(ZL + R) = Va - Vb$

$I1 * (R + Zc) - I2 * R = Vc - Vb$

$$I1 = \frac{\begin{vmatrix} Va - Vb & -(ZL + R) \\ Vc - Vb & -R \end{vmatrix}}{\begin{vmatrix} R & -(ZL + R) \\ R + Zc & -R \end{vmatrix}}$$

$$I2 = \frac{\begin{vmatrix} R & Va - Vb \\ R + Zc & Vc - Vb \end{vmatrix}}{\begin{vmatrix} R & -(ZL + R) \\ R + Zc & -R \end{vmatrix}}$$

$I1 = 9.332 - 9.363i$ $I2 = -12.982 - 3.246i$

$VAN = -I2 * ZL = -15.296 + 61.175i$

$|VAN| = 63.058$ $\beta 1 = 104.038$

$VBN = R * (I2 - I1) = -243.142 + 61.175i$

$|VBN| = 231.376$ $\beta 2 = 164.669$

$VCN = Zc * I1 = -119.219 - 118.825i$

$|VCN| = 168.323$ $\beta 3 = -135.095$

$Sa = |I2|^2 * ZL = 843.809i$

$Sb = |I2 - I1|^2 * R = 5.353 * 10^3$

$Sc = |I1|^2 * Zc = -2.225i * 10^3$

$S = Sa + Sb + Sc = (5.353 - 1.381i) * 10^3$

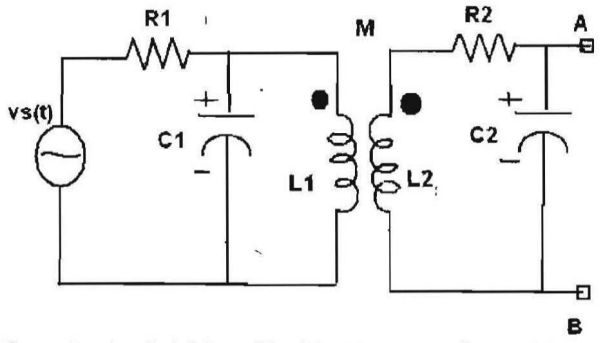
Q-3)

For the circuit, unknown values are: V_s(t)=100 cos(5t) ,$R_1 = 12\ \Omega$, $R_2 = 5\ \Omega$, $L_1 = 0.3\ H$,$L_2 = 1.2\ H$,$C_1 = C_2 = \frac{1}{100} F$ and $M = 0.6\ H$.

a) Calculate the Thévenin Equivalent Circuit between A-B port.

b) What is the impedance value which satisfy the max. power between A-B.

c)For the impedance value found in b, calculate the apparent power.



$$X_{L1} = j * w * L1 = 1.5i$$

$$X_{L2} = j * w * L2 = 6i$$

$$X_{C1} = X_{c2} = \frac{1}{i * w * C1} = -20i$$

a) $V_s = R1 * I_1 + X_{C1} * (I_1 - I_2)$

$0 = X_{L1} * I_2 + X_M * I_3 - X_{C1}(I_2 - I_1)$

$0 = X_{L2} * I_3 + X_M * I_2 + R2 * I_3 - X_{C2} * I_3$

$100 = 12 * I_1 - 20i * (I_1 - I_2)$

$0 = 1.5i * I_2 + 3i * I_3 + 20(I_2 - I_1)$

$0 = 6i * I_3 + 3i * I_2 + 5 * I_3 + 20i * I_3$

$$\begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 12-20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5+26i \end{bmatrix} * \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

$$I_1 = \frac{\left\| \begin{bmatrix} 100 & 20i & 0 \\ 0 & 21.5i & 3i \\ 0 & 3i & 5+26i \end{bmatrix} \right\|}{\left\| \begin{bmatrix} 12-20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5+26i \end{bmatrix} \right\|} = 8.303 + 0.766i$$

$$I_2 = \frac{\left\| \begin{bmatrix} 12-20i & 100 & 0 \\ -20i & 0 & 3i \\ 0 & 0 & 5+26i \end{bmatrix} \right\|}{\left\| \begin{bmatrix} 12-20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5+26i \end{bmatrix} \right\|} = 7.843 + 0.748i$$

$$I_3 = \frac{\left\| \begin{bmatrix} 12-20i & 20i & 100 \\ -20i & 21.5i & 0 \\ 0 & 3i & 0 \end{bmatrix} \right\|}{\left\| \begin{bmatrix} 12-20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5+26i \end{bmatrix} \right\|} = -0.857 - 0.25i$$

$$Voc = X_{c2} * (-I_3) = 5.021 - 17.134i$$

AB kısa devre olursa

$$100 = 12 * I_1 - 20i * (I_1 - I_2)$$
$$0 = 1.5i * I_2 + 3i * I_3 + 20(I_2 - I_1)$$
$$0 = 6i * I_3 + 3i * I_2 + 5 * I_3$$

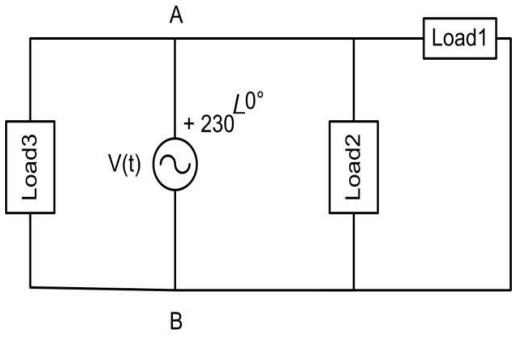
$$\begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 12 - 20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5 + 6i \end{bmatrix} * \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

$$I_{sc} = \frac{\left\| \begin{bmatrix} 12 - 20i & 20i & 100 \\ -20i & 21.5i & 0 \\ 0 & 3i & 0 \end{bmatrix} \right\|}{\left\| \begin{bmatrix} 12 - 20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5 + 6i \end{bmatrix} \right\|} = -2.326 - 2.326i$$

$$Zth = \frac{Voc}{Isc} = 2.604 + 4.763i$$
$$Zmax = Zth^* = -2.604 - 4.763i$$
$$S = Zmax * Isc^2 = 51.521 + 28.17i$$

Q-4)

The source $V(t) = 230\angle 0°$ feeds 3 parallel Loads. For the loads below
Load 1 : S=25 kVA, PF=0.6 inductive,
Load 2 :P=5 kW, PF=0.8 capacitive
Load 3 :P=10 kW, PF=1
a) Calculate the condansator value to set the power factor at 0.95. (Freq.=50 Hz)
b) Find the current value before and after the compansation. Explain the difference, if it exist.



$$P1 = S * Pf = 25 * 0.6 = 15$$

$$Q1 = S * \sin(\arccos(Pf)) = 25 * \sin(\arccos(0.6)) = 20$$

$$P2 = 5$$

$$Q2 = -P * \frac{\sin(\arccos(Pf))}{Pf} = -5 * \sin\frac{(\arccos(0.8))}{0.8} = -3.75$$

$$P3 = 10$$

$$Q3 = P * \frac{\sin(\arccos(Pf))}{Pf} = 10 * \frac{\sin(\arccos(1))}{1} = 0$$

a) Kompanzasyondan önce toplam reaktif güç

$$gf = \frac{P1 + P2 + P3}{\sqrt{(P1 + P2 + P3)^2 + (Q1 + Q2 + Q3)^2}} = 0.879$$

$$Qtoplam_{ilk} = Q1 + Q2 + Q3 = 20 - 3.75 + 0 = 16.25$$

Güç faktörünü 0.95 yapmak için gereken kondansatör büyüklüğü

$$Qtoplam_{son} = (P1 + P2 + P3) * \tan\big(acos(0.95)\big) = 9.861$$

$$Q_C = Qtoplam_{son} - Qtoplam_{ilk} = 16.25 - 9.861 = 6.389$$

$$X_c = \frac{V^2}{Q_c} = 8.279 * 10^3$$

$$c = \frac{1}{2 * \pi * f * X_c} = 3.845 * 10^{-4}$$

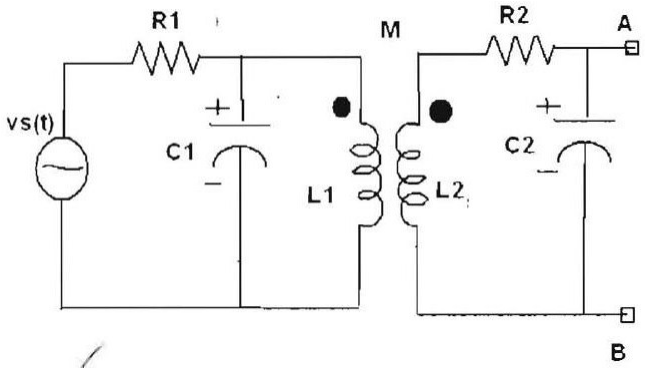
b) $S_{ilk} = \sqrt{(P1 + P2 + P3)^2 + Qtoplam_{ilk}{}^2} = 34.118$

$$I_{ilk} = \frac{S_{ilk}}{V} = 0.148$$

$$S_{son} = \sqrt{(P1 + P2 + P3)^2 + Qtoplam_{son}{}^2} = 31.579$$

Q-5) Voltage source Vs(t)=100.Cos(5t) and R1=12 ohm, R2=5 ohm, L1=0.3h, L2=1.2H, C1=C2=1/100 F and mutual inductance (M) is 0.6 H.

A) Please find thevenin equivalent between A and B.
B) To draw maximum power from voltage source, please find the impedance that must be connected between A and B.



$$X_{L1} = j * w * L1 = 1.5i$$

$$X_{L2} = j * w * L2 = 6i$$

$$X_{C1} = X_{c2} = \frac{1}{i * w * C1} = -20i$$

a) $V_s = R1 * I_1 + X_{C1} * (I_1 - I_2)$
$0 = X_{L1} * I_2 + X_M * I_3 - X_{C1}(I_2 - I_1)$
$0 = X_{L2} * I_3 + X_M * I_2 + R2 * I_3 - X_{C2} * I_3$
$100 = 12 * I_1 - 20i * (I_1 - I_2)$

$0 = 1.5i * I_2 + 3i * I_3 + 20(I_2 - I_1)$

$0 = 6i * I_3 + 3i * I_2 + 5 * I_3 + 20i * I_3$

$$\begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 12-20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5+26i \end{bmatrix} * \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

$$I_1 = \frac{\left\| \begin{bmatrix} 100 & 20i & 0 \\ 0 & 21.5i & 3i \\ 0 & 3i & 5+26i \end{bmatrix} \right\|}{\left\| \begin{bmatrix} 12-20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5+26i \end{bmatrix} \right\|} = 8.303 + 0.766i$$

$$I_2 = \frac{\left\| \begin{bmatrix} 12-20i & 100 & 0 \\ -20i & 0 & 3i \\ 0 & 0 & 5+26i \end{bmatrix} \right\|}{\left\| \begin{bmatrix} 12-20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5+26i \end{bmatrix} \right\|} = 7.843 + 0.748i$$

$$I_3 = \frac{\left|\begin{bmatrix} 12-20i & 20i & 100 \\ -20i & 21.5i & 0 \\ 0 & 3i & 0 \end{bmatrix}\right|}{\left|\begin{bmatrix} 12-20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5+26i \end{bmatrix}\right|} = -0.857 - 0.25i$$

$$Voc = X_{c2} * (-I_3) = 5.021 - 17.134i$$

AB kısa devre olursa

$$100 = 12 * I_1 - 20i * (I_1 - I_2)$$
$$0 = 1.5i * I_2 + 3i * I_3 + 20(I_2 - I_1)$$
$$0 = 6i * I_3 + 3i * I_2 + 5 * I_3$$

$$\begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 12-20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5+6i \end{bmatrix} * \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

$$I_{sc} = \frac{\left|\begin{bmatrix} 12-20i & 20i & 100 \\ -20i & 21.5i & 0 \\ 0 & 3i & 0 \end{bmatrix}\right|}{\left|\begin{bmatrix} 12-20i & 20i & 0 \\ -20i & 21.5i & 3i \\ 0 & 3i & 5+6i \end{bmatrix}\right|} = -2.326 - 2.326i$$

$$Zth = \frac{Voc}{Isc} = 2.604 + 4.763i$$
$$Zmax = Zth^* = -2.604 - 4.763i$$

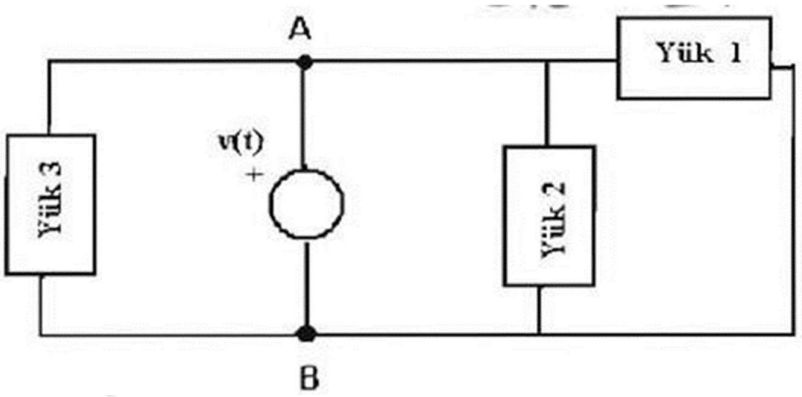
Q-6) The source whose voltage is $230\ V$ is providing power to 3 loads which are Yük-1, Yük-2 and Yük-3:

Yük 1: S= 20 kVA, pf=0.6 (inductive)

Yük 2: P= 10 kW, pf=0.8 (capacitive)

Yük 3: P= 10 kW, pf=0.6 (inductive)

A) Please find the capacity value of the capacitor to make power factor 0.95 (f=50hz). The location of the capacity must be in between A and B points.
B) Please find the current given by source before and after compensation.



$S1 := 20000 \quad P1 := S1 \cdot 0.6 \quad Q1 := S1 \cdot \sin(\acos(0.6)) \qquad P1 = 12000 \quad Q1 = 16000$

$P2 := 10000 \quad S2 := \frac{P2}{0.8} \quad Q2 := -S2 \cdot \sin(\acos(0.8)) \qquad P2 = 10000 \quad Q2 = -7500$

$S2 = 12500$

$P3 := 10000 \quad S3 := \frac{P3}{0.6} \quad Q3 := S3 \cdot \sin(\acos(0.6)) \qquad P3 = 10000 \quad Q3 = 13333.333$

$S3 = 16667$

TOPLAM GÜÇ

$PT := P1 + P2 + P3 \qquad QT := Q1 + Q2 + Q3$

$ST := (PT) + i \cdot QT \qquad ST = 32000 + 21833.333i$

$|ST| = 38739$

$gf := \frac{PT}{|ST|}$

$gf = 0.826$

$Qko := Q1 + Q2 + Q3$ KOMPANZASYON ONCESI Q

$Qko = 2.183 \times 10^4$

$Qks := (P1 + P2 + P3) \cdot \tan(\acos(0.95))$ KOMPANZASYON SONRASI Q

$Qks = 1.052 \times 10^4$

$Qc := Qko - Qks$ GEREKLI KAPASITE

$Qc = 1.132 \times 10^4$

$Xc := \frac{230^2}{Qc}$

$Xc = 4.675$

$C := \frac{1}{2 \cdot \pi \cdot 50 Xc}$

$C = 6.809 \times 10^{-4}$

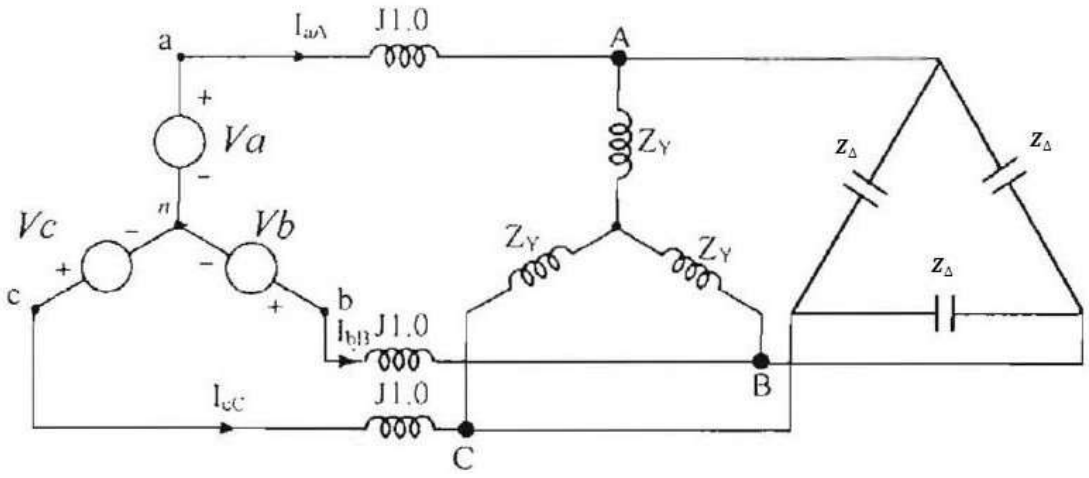
$Sko := \sqrt{(P1 + P2 + P3)^2 + Qko^2} \qquad Sks := \sqrt{(P1 + P2 + P3)^2 + Qks^2}$

Iko=Sko/230=168.430 A Iks=Sks/230=146.45A__ko:kompanzasyon öncesi___ks:kompanzasyon sonrası

Q-7) Phase voltages are $V_a = 220\angle 30^\circ$, $V_b = 220\angle 150^\circ$ and $V_a = 220\angle -90^\circ$. The loads information: $Z_\Delta = -j100ohm$ and $Z_\gamma = j100ohm$.

A) Please find IaA, IbB and IcC.

B) Please find total active, reactive and apperant power consumed by loads.



Sistem dengelidir

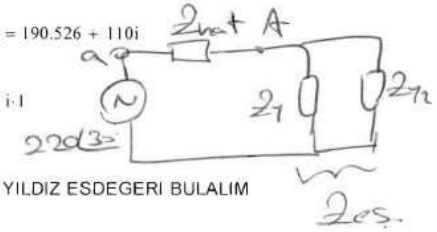
Nedeni :

Kaynak tarafýnda her faz geriliminin rms deðerleri eþit ve aralarýnda 120 þer derece fark vardir. Yük tarafinda ise her faza bagli empedanslar birbirine esittir. Sistemi yildiz a cevirerek cozmek en kolay yol

a)

$$Va := 220\cdot\left(\cos\left(30\cdot\frac{\pi}{180}\right) + i\cdot\sin\left(30\cdot\frac{\pi}{180}\right)\right) \qquad Va = 190.526 + 110i$$

$$Z\Delta := -i\cdot 100 \qquad ZY := 100i \qquad Zhat := i\cdot 1$$



$$ZY2 := \frac{Z\Delta}{3} \qquad ZY2 = -33.333i$$ UCGEN YUKUN YILDIZ ESDEGERI BULALIM

$$Zes := \frac{ZY\cdot ZY2}{ZY + ZY2}$$ TOPLAM YUK $\qquad Zes = -50i$

$$IaA := \frac{Va}{Zhat + Zes} \qquad IaA = -2.245 + 3.888i$$ kaynaktan çekilen akım

4,489 (120 A

$$IbB := |IaA|\cdot\left(\cos\left(\arg(IaA) - \frac{4\pi}{3}\right) + i\cdot\sin\left(\arg(IaA) - \frac{4\pi}{3}\right)\right)$$

$$IcC := |IaA|\cdot\left(\cos\left(\arg(IaA) - \frac{4\pi}{3}\right) + i\cdot\sin\left(\arg(IaA) - \frac{4\pi}{3}\right)\right)$$

b)

$$VA := Va - Zhat\cdot IaA \qquad VA = 194.414 + 112.245i$$

$$SA := VA\cdot\overline{IaA}$$

$$SA = -1.008i\times 10^3 \qquad S := 3\cdot SA$$

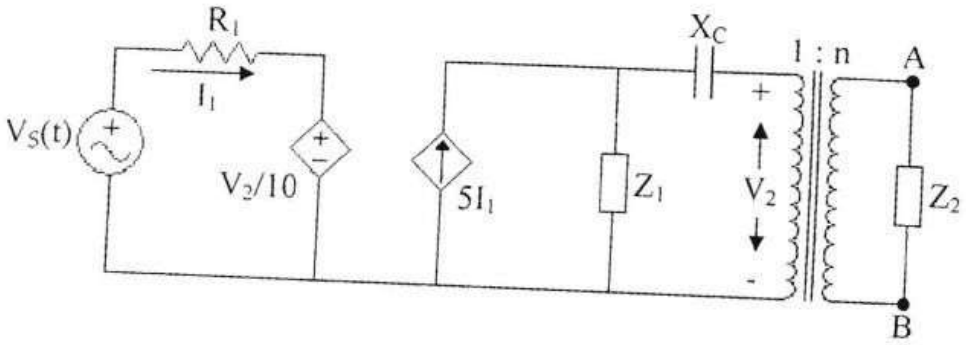
$$P := Re(S) \qquad Q := Im(S)$$

$$S = -3.024i\times 10^3$$

$$P = 0$$

$$Q = -3.024\times 10^3$$

Q-8) To draw maximum power from the circuit, please find Z_2 value. $V_s(t) = 20\cos 100t$, $R_l = 20ohm$, $Z_l = 10 + 10j$, $X_c = -j$, n=5.



Maksimum güç transferi için Z2=Zth olmalıdır. Voc ve Isc is Rth'i bulacağız.

Voc için:

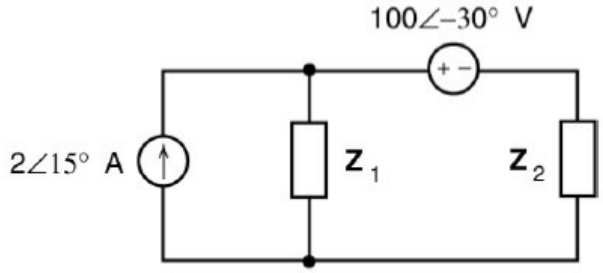
Vs=I1.R1+V2/10
V2=Z1.5I1 (Voc koşulunda Z2üzerinden akım geçmez)

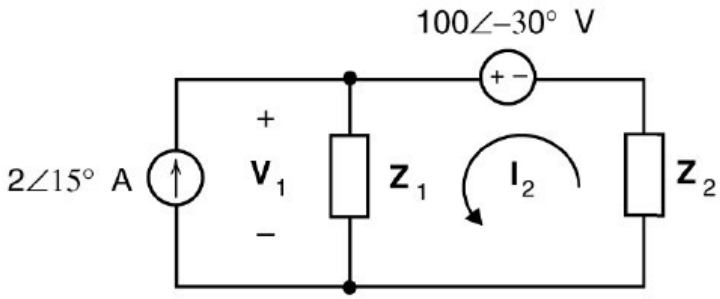
20=I1.20+ Z1.5I1/10
20=20.I1 + (10+10i).5.I1/10
I1=20/25+5i=0.769-0.154i
Voc=46.15+30.75i

Isc için:
V2=0 ise Vs=I1R1, I1=1A
Isc=5.I1 Z1/(Z1+Xc)=5.((10+10i)/(10+10i-j))= 5.2486 + 0.2762i
Zth=Vos/Isc=(46.15+30.75i)/(5.2486 + 0.2762i)= 9.0760 + 5.3811i

Z2=conjugate(Zth).25=226.90 + 134.53i

Q-9) The current source the circuit shown in Figure below supplies 131.16 – *j*36.048 VA and the voltage source supplies 64.2275 – 87.8481VA. Determine the values of the impedances Z1 and Z2.





$i := \sqrt{-1}$

The voltage of current source VIs:

$SIs := 131.16 - i \cdot 36.048$

$Is := 2 \cdot \left(\cos\left(\frac{15 \cdot \pi}{180} \right) + i \cdot \sin\left(\frac{15 \cdot \pi}{180} \right) \right)$

$Is = 1.932 + 0.518i$

$VIs := \frac{SIs}{\overline{Is}} = 68.01 - 0.436i$

$|VIs| = 68.012 \qquad \arg(VIs) = -0.368\text{deg}$

The current of voltage source IVs:

$SVs := 64.2275 - i \cdot 87.8481$

$Vs := 100 \cdot \left(\cos\left(\frac{-30 \cdot \pi}{180} \right) + i \cdot \sin\left(\frac{-30 \cdot \pi}{180} \right) \right)$

$IVs := \frac{\overline{SVs}}{\overline{Vs}} = 0.995 + 0.44i$

$|IVs| = 1.088$

$\arg(IVs) = 23.829\text{deg}$

The current I2 :

$I2 := IVs = 0.995 + 0.44i$

the voltage of Z2 VZ2: KVL to 2.loop

$VZ2 := Vs - VIs = 18.592 - 49.564i$

$|VZ2| = 52.936 \qquad \arg(VZ2) = -69.438\text{deg}$

Z2 impedance :

$Z2 := \frac{VZ2}{I2} = -2.772 - 48.565i$

$|Z2| = 48.644 \qquad \arg(Z2) = -93.267\text{deg}$

KCL at the top of node :

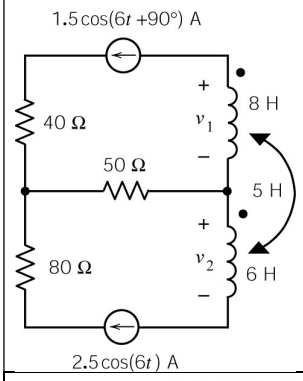
$I1 := Is - I2 = 0.936 + 0.078i$

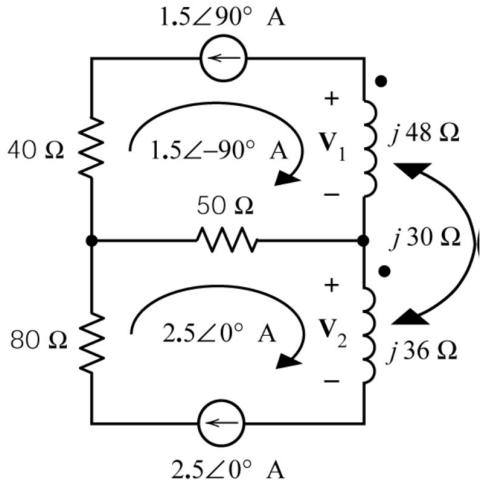
Z1 impedance :

$Z1 := \frac{VIs}{I1} = 72.092 - 6.47i$

$|Z1| = 72.382 \qquad \arg(Z1) = -5.129\text{deg}$

Q-10) The circuit shown in Figure below is represented in the time domain. Determine coil voltages *v*1 and *v*2.





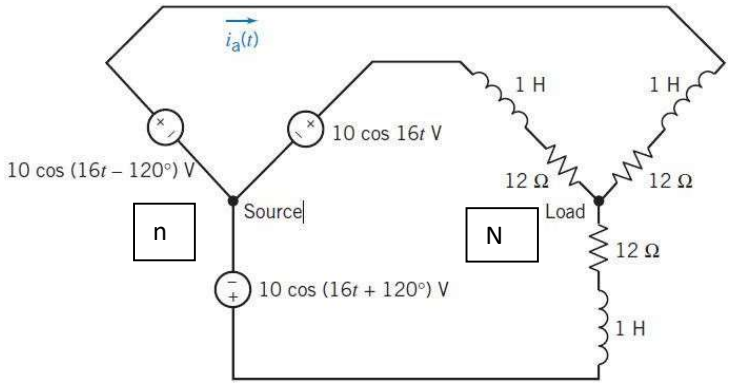
$V1 = j48(1.5\angle -90\text{deg}) + 30j(2.5\angle 0\text{deg})$ =$103.97\angle 46.17\text{deg}$

$V2 = j36(2.5\angle -90\text{deg}) + 30j(2.5\angle -90\text{deg})$ =$100.62\angle 63.63\text{deg}$

Q-11) Wye-connected source and load are shown in Figure below.

(a) Determine the rms value of the current *i*A (*t*).

(b) Determine the average power delivered to the load.



Sistem dengelidir.
Va=7.07∠0° , Vb=V = 7.07∠−120° , Vc = 7.07∠120°

Za=Zb=Zc=12 + j16 Ω

VnN=(Za.Zc.Vb+Za.Zb.Vc+Zb.Z.Va)/(Za.Zc+Za.Zb+Zb.Zc)

IaA=(Va-VnN)/Za=0.212-0.238i (line curent)

Sa=conjugate(IaA).IaA.Za=1.5+2i

Stotal=3.Sa=4.5+6i

A) 0.354 amper (rms)
B) 4.5 watt

Q-12) The circuit in Figure below consists of a source connected to a load.

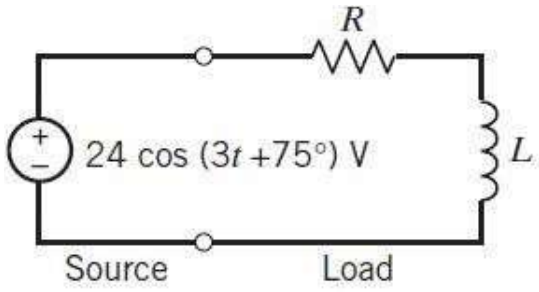
(a) Suppose R = 9 Ω and L = 5 H. Determine the average, complex, and reactive powers delivered by the source to the load.

(b) Suppose R = 15 Ω and L = 3 H. Determine the average, complex, and reactive powers delivered by the source to the load.

(c) Suppose the source delivers 8.47 + j14.12 VA to the load. Determine the values of the resistance, R, and the inductance, L.

(d) Suppose the source delivers 14.12 + j8.47 VA to the load. Determine the values of the

resistance, R, and the inductance, L.



A) Zl=9+15i
Ia=(24∠75deg)/Zl=1.3191 + 0.3773i
S_source_a=conjugate(Ia)* (24∠75deg)=16.9404 +28.2360i

B) Zl=15+9i
Ia= 0.9863 + 0.9537i
S_source_a= -15.9823 +28.7887i

C) Ia=conj(S/Vs)= 0.6596 +0.1886i
Zl=Vs/Ia=17.9954 +30.0004i ise R=18 ve L=10'dur.

D) C durumun tam tersi R ve L'nin değerinin yer değiştirmiş hali için çözüm olacak.
R=30 ve L=6^dır.